

Research interests

My current research focuses on the effect of extra scalar degrees of freedom on our universe. On large scales, I have studied their impact on the Hubble tension and on the CMB visibility function through their collapse into compact objects. On short scales, I have explored fifth forces and screening mechanisms originating from non-minimal couplings to gravity, developing a code (**FeynMG**) that helps test them using collider data.

Current position

2025- **Postdoctoral Research Fellow** at **University of Pennsylvania, USA**.

Collaborators: Prof. Mark Trodden and Prof. Justin Khoury.

Education

2023-2025 **Postdoctoral Research Associate** at IPPP, **Durham University, UK**.

Collaborators: Dr. Djuna Croon and Dr. Ivan Martinez-Soler.

2020-2023 **PhD in Particle Cosmology** at **University of Nottingham, UK**.

Title: *FeynMG: Automating particle physics calculations in scalar-tensor theories*.

Supervisor: Prof. Edmund J. Copeland and Dr. Pete Millington.

2019-2020 **Part III** in Applied Mathematics and Theoretical Physics at **Univeristy of Cambridge, UK**.

Essay: *Tunneling Transitions in Quantum Mechanics, Field Theory and Gravity*.

Supervisor: Prof. Fernando Quevedo.

2016-2019 **B.Sc** in Physics and Theoretical Physics at **University of Nottingham, UK**.

Third-year project: *Scalar Fields in Cosmology and the Swampland Conjecture*.

Supervisor: Prof. Edmund J. Copeland.

Selected talks

Title: **Updated constraints on the primordial power spectrum at sub-Mpc scales**

- From Cosmos to Lab, Mainz (Sep 2025)
- PASCOS'25, Durham (Jun 2024)
- News from Dark, Montpellier (Sep 2025)

Title: **Screening mechanisms from pa particle's perspective**

- King's College London (Feb 2025)
- UKCosmo, Nottingham (Apr 2025)
- University of Sheffield (Nov 2024)
- Liverpool University (Nov 2024)

Title: **CMB bounds on accreting Extended Dark Matter Objects**

- DMLAND, MITP, Mainz (Sep 2024)
- University of Nottingham (Jun 2024)
- Beyond WIMPS, Durham (Mar 2024)
- Cosmology from home (Jun 2024)

Title: **How to study modified gravity as a particle theory and not collapse in the process**

- Perimeter Institute, Canada (Jul 2024)
- UKCosmo, Cambridge (May 2023)
- COSMOS'23, IFT, Madrid (Sep 2023)
- PASCOS'23, California (Jul 2023)

Title: **Addressing the Hubble tension with scalar fields**

- UNAM, Mexico (Apr 2024)
- University of Nottingham (Dec 2022)

Computing skills

I have developed the following codes:

- MATHEMATICA:** **FeynMG:** A Feynrules subpackage for studying scalar-tensor theories within particle theory pipelines.
CMB accretion: A numerical code to predict and constrain the influence of Extended Dark Matter objects on the CMB visibility function.
- PYTHON:** **EDOBounds:** Repository for constraints on Extended Dark Matter Objects, allowing the plotting of various bounds combinations for any given shape or radius.

Teaching experience

- 2024-2025 Tutor for master's level modules: '*Introduction to Field Theory*' and '*Cosmology*'.
- 2023-2024 Tutor for first year physics course '*Foundations of physics*' at Durham University.
- 2020-2023 Workshop demonstrator for '*Quantum dynamics*', '*Computing*', '*Symmetries and action principles*', '*Fourier analysis*' and '*Atoms, photons and fundamental particles*' at University of Nottingham.
- 2017-2022 Physics and Mathematics private tutor for A-level students.

Academic service

- 2024 - Interviewer for EuCAPT series of videos (started in September 2024).
- 2024 - Astroparticle Journal Club organiser at IPPP, Durham University.
- 2023 - IPPP Postdoctoral Representative at Staff Committee Meetings in the Physics Department.
- 2022 - Referee for Physical Review Journal D and European Physical Journal C.
- 2022 - 2023 Web page editor for UK Cosmo.
- 2021 - 2023 Web page editor for the Particle Cosmology group at University of Nottingham, UK.
- 2021 - 2022 Coordinator of the '*Particle Cosmology Student Journal Clubs*', University of Nottingham, UK.

Outreach Activities

- Sep 2024 1.5 hour podcast on "Modified Gravity and Cosmology", in Physics for Students YouTube channel.
- May 2024 20-min talk titled: "The expansion of the universe and the Hubble tension", Pint of Science, Durham
- Mar 2024 40-min talk titled: "The expansion of the universe and the Hubble tension", Café Scientifique, Durham
- 2018- Regularly upload videos and do lives on **TikTok** about physics and academia, with a total of 11K followers and ~ 100K likes.

Awards and Scholarships

- Feb 2025 **Seal of Excellence** from the European Commission Horizon Europe for my project proposal FICoSSMe.
- Jun 2023 Paul Dirac prize at Erice International School of Subnuclear Physics for my contributions and special talent talk.
- May 2023 Andrew Hendry Scholarship Endowed Award 2023 for my PhD trajectory.
- Mar 2022 1st Prize in the 2022 Physics and Astronomy Poster competition at University of Nottingham, UK.
- Jan 2017 Sir Peter Mansfield Award for excellent academic results at University of Nottingham, UK.

Official visits

- Jun 2024 2-week visit to the Perimeter Institute, Waterloo, Canada.
Collaborator: Prof. Cliff Burgess.
- Mar 2024 1-week visit to the Physics department of the University of Manchester, UK.
Collaborator: Dr. Peter Millington.
- Jun 2022 1-week visit to the Physics department of the University of Glasgow, UK.
Collaborator: Prof. Christoph Englert.